

Epoch 100: train_mse=0.24750, test_mse=0.24759, test_accuracy=0.5557
Epoch 200: train_mse=0.24682, test_mse=0.24697, test_accuracy=0.5557
Epoch 300: train_mse=0.24663, test_mse=0.24680, test_accuracy=0.5557
Epoch 400: train_mse=0.24653, test_mse=0.24671, test_accuracy=0.5557
Epoch 500: train_mse=0.24645, test_mse=0.24664, test_accuracy=0.5557
Epoch 600: train_mse=0.24637, test_mse=0.24657, test_accuracy=0.5557
Epoch 700: train_mse=0.24629, test_mse=0.24650, test_accuracy=0.5557
Epoch 800: train_mse=0.24621, test_mse=0.24642, test_accuracy=0.5557
Epoch 900: train_mse=0.24612, test_mse=0.24634, test_accuracy=0.5557
Epoch 1000: train_mse=0.24602, test_mse=0.24625, test_accuracy=0.5557
Epoch 1100: train_mse=0.24592, test_mse=0.24616, test_accuracy=0.5557
Epoch 1200: train_mse=0.24581, test_mse=0.24606, test_accuracy=0.5557
Epoch 1300: train_mse=0.24569, test_mse=0.24594, test_accuracy=0.5557
Epoch 1400: train_mse=0.24556, test_mse=0.24582, test_accuracy=0.5557
Epoch 1500: train_mse=0.24542, test_mse=0.24569, test_accuracy=0.5557
Epoch 1600: train_mse=0.24526, test_mse=0.24555, test_accuracy=0.5557
Epoch 1700: train_mse=0.24509, test_mse=0.24539, test_accuracy=0.5557
Epoch 1800: train_mse=0.24491, test_mse=0.24521, test_accuracy=0.5557
Epoch 1900: train_mse=0.24470, test_mse=0.24502, test_accuracy=0.5557
Epoch 2000: train_mse=0.24447, test_mse=0.24481, test_accuracy=0.5557
Epoch 2100: train_mse=0.24423, test_mse=0.24458, test_accuracy=0.5557
Epoch 2200: train_mse=0.24396, test_mse=0.24433, test_accuracy=0.5557
Epoch 2300: train_mse=0.24366, test_mse=0.24405, test_accuracy=0.5557
Epoch 2400: train_mse=0.24334, test_mse=0.24375, test_accuracy=0.5557
Epoch 2500: train_mse=0.24299, test_mse=0.24342, test_accuracy=0.5557
Epoch 2600: train_mse=0.24260, test_mse=0.24306, test_accuracy=0.5557
Epoch 2700: train_mse=0.24218, test_mse=0.24267, test_accuracy=0.5557

Epoch 2800: train_mse=0.24173, test_mse=0.24225, test_accuracy=0.5557
Epoch 2900: train_mse=0.24124, test_mse=0.24179, test_accuracy=0.5557
Epoch 3000: train_mse=0.24071, test_mse=0.24129, test_accuracy=0.5557
Epoch 3100: train_mse=0.24015, test_mse=0.24076, test_accuracy=0.5557
Epoch 3200: train_mse=0.23954, test_mse=0.24019, test_accuracy=0.5557
Epoch 3300: train_mse=0.23889, test_mse=0.23959, test_accuracy=0.5558
Epoch 3400: train_mse=0.23820, test_mse=0.23894, test_accuracy=0.5558
Epoch 3500: train_mse=0.23747, test_mse=0.23826, test_accuracy=0.5558
Epoch 3600: train_mse=0.23670, test_mse=0.23754, test_accuracy=0.5560
Epoch 3700: train_mse=0.23589, test_mse=0.23678, test_accuracy=0.5558
Epoch 3800: train_mse=0.23504, test_mse=0.23599, test_accuracy=0.5571
Epoch 3900: train_mse=0.23416, test_mse=0.23517, test_accuracy=0.5586
Epoch 4000: train_mse=0.23324, test_mse=0.23432, test_accuracy=0.5706
Epoch 4100: train_mse=0.23230, test_mse=0.23344, test_accuracy=0.6326
Epoch 4200: train_mse=0.23133, test_mse=0.23254, test_accuracy=0.6701
Epoch 4300: train_mse=0.23035, test_mse=0.23162, test_accuracy=0.6928
Epoch 4400: train_mse=0.22934, test_mse=0.23070, test_accuracy=0.6937
Epoch 4500: train_mse=0.22833, test_mse=0.22976, test_accuracy=0.6941
Epoch 4600: train_mse=0.22731, test_mse=0.22882, test_accuracy=0.6941
Epoch 4700: train_mse=0.22629, test_mse=0.22788, test_accuracy=0.6941
Epoch 4800: train_mse=0.22528, test_mse=0.22694, test_accuracy=0.6941
Epoch 4900: train_mse=0.22427, test_mse=0.22602, test_accuracy=0.6941
Epoch 5000: train_mse=0.22328, test_mse=0.22511, test_accuracy=0.6941
Epoch 5100: train_mse=0.22231, test_mse=0.22422, test_accuracy=0.6941
Epoch 5200: train_mse=0.22136, test_mse=0.22335, test_accuracy=0.6941
Epoch 5300: train_mse=0.22043, test_mse=0.22251, test_accuracy=0.6941
Epoch 5400: train_mse=0.21954, test_mse=0.22171, test_accuracy=0.6939

Epoch 5500: train_mse=0.21868, test_mse=0.22093, test_accuracy=0.6941
Epoch 5600: train_mse=0.21785, test_mse=0.22019, test_accuracy=0.6941
Epoch 5700: train_mse=0.21707, test_mse=0.21948, test_accuracy=0.6941
Epoch 5800: train_mse=0.21632, test_mse=0.21881, test_accuracy=0.6941
Epoch 5900: train_mse=0.21560, test_mse=0.21818, test_accuracy=0.6944
Epoch 6000: train_mse=0.21493, test_mse=0.21758, test_accuracy=0.6943
Epoch 6100: train_mse=0.21430, test_mse=0.21702, test_accuracy=0.6946
Epoch 6200: train_mse=0.21370, test_mse=0.21650, test_accuracy=0.6946
Epoch 6300: train_mse=0.21314, test_mse=0.21602, test_accuracy=0.6944
Epoch 6400: train_mse=0.21262, test_mse=0.21557, test_accuracy=0.6944
Epoch 6500: train_mse=0.21214, test_mse=0.21515, test_accuracy=0.6944
Epoch 6600: train_mse=0.21169, test_mse=0.21476, test_accuracy=0.6941
Epoch 6700: train_mse=0.21127, test_mse=0.21441, test_accuracy=0.6941
Epoch 6800: train_mse=0.21088, test_mse=0.21408, test_accuracy=0.6943
Epoch 6900: train_mse=0.21052, test_mse=0.21378, test_accuracy=0.6943
Epoch 7000: train_mse=0.21019, test_mse=0.21351, test_accuracy=0.6943
Epoch 7100: train_mse=0.20989, test_mse=0.21326, test_accuracy=0.6943
Epoch 7200: train_mse=0.20961, test_mse=0.21303, test_accuracy=0.6943
Epoch 7300: train_mse=0.20935, test_mse=0.21282, test_accuracy=0.6943
Epoch 7400: train_mse=0.20911, test_mse=0.21264, test_accuracy=0.6943
Epoch 7500: train_mse=0.20889, test_mse=0.21246, test_accuracy=0.6943
Epoch 7600: train_mse=0.20869, test_mse=0.21231, test_accuracy=0.6943
Epoch 7700: train_mse=0.20851, test_mse=0.21217, test_accuracy=0.6943
Epoch 7800: train_mse=0.20834, test_mse=0.21204, test_accuracy=0.6943
Epoch 7900: train_mse=0.20819, test_mse=0.21193, test_accuracy=0.6944
Epoch 8000: train_mse=0.20805, test_mse=0.21182, test_accuracy=0.6944
Epoch 8100: train_mse=0.20792, test_mse=0.21173, test_accuracy=0.6944

Epoch 8200: train_mse=0.20780, test_mse=0.21164, test_accuracy=0.6944
Epoch 8300: train_mse=0.20770, test_mse=0.21157, test_accuracy=0.6944
Epoch 8400: train_mse=0.20760, test_mse=0.21150, test_accuracy=0.6944
Epoch 8500: train_mse=0.20751, test_mse=0.21144, test_accuracy=0.6944
Epoch 8600: train_mse=0.20742, test_mse=0.21138, test_accuracy=0.6944
Epoch 8700: train_mse=0.20735, test_mse=0.21133, test_accuracy=0.6944
Epoch 8800: train_mse=0.20728, test_mse=0.21129, test_accuracy=0.6944
Epoch 8900: train_mse=0.20722, test_mse=0.21125, test_accuracy=0.6944
Epoch 9000: train_mse=0.20716, test_mse=0.21121, test_accuracy=0.6944
Epoch 9100: train_mse=0.20711, test_mse=0.21118, test_accuracy=0.6944
Epoch 9200: train_mse=0.20706, test_mse=0.21115, test_accuracy=0.6944
Epoch 9300: train_mse=0.20701, test_mse=0.21113, test_accuracy=0.6944
Epoch 9400: train_mse=0.20697, test_mse=0.21111, test_accuracy=0.6944
Epoch 9500: train_mse=0.20693, test_mse=0.21109, test_accuracy=0.6944
Epoch 9600: train_mse=0.20690, test_mse=0.21107, test_accuracy=0.6944
Epoch 9700: train_mse=0.20687, test_mse=0.21105, test_accuracy=0.6944
Epoch 9800: train_mse=0.20684, test_mse=0.21104, test_accuracy=0.6946
Epoch 9900: train_mse=0.20681, test_mse=0.21103, test_accuracy=0.6946
Epoch 10000: train_mse=0.20679, test_mse=0.21102, test_accuracy=0.6946

Best epoch: 10000

Best test MSE: 0.21102

Final training accuracy: 0.7052

Final test accuracy: 0.6946

Rows used after filtering: 28387

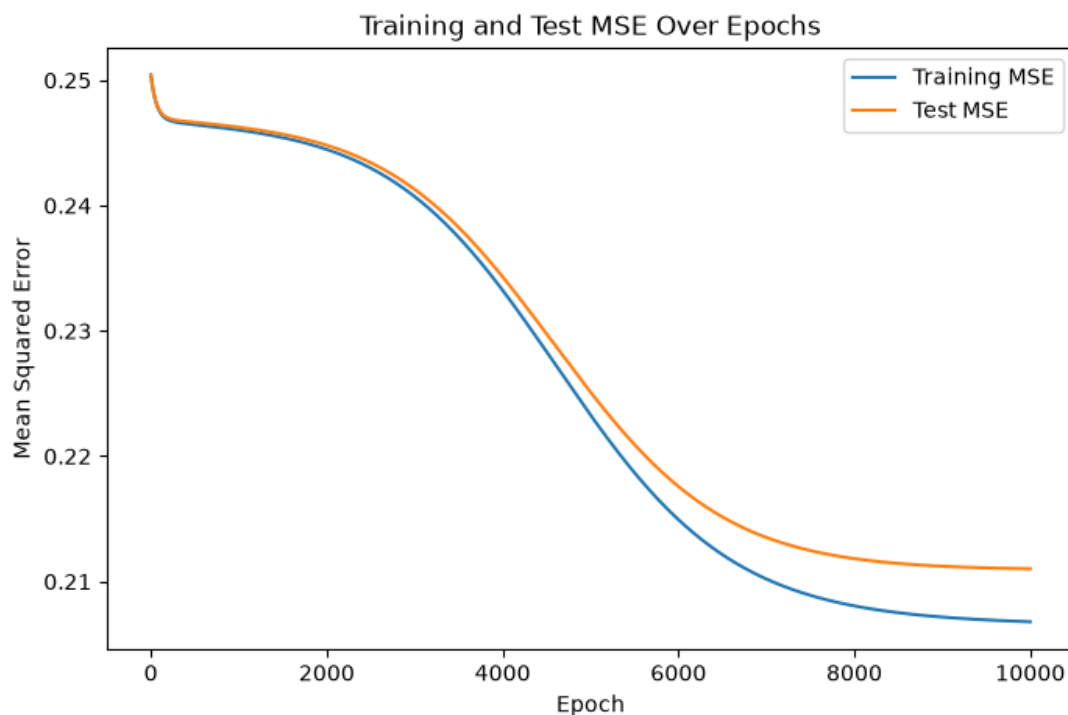
Training set size: 22709

Test set size: 5678

Trained for the full 10,000 epoch (never triggered early stopping, test MSE was still slowly improving at the end). Final numbers:

- Best epoch: 10000
- Best test MSE: 0.21102
- Final training accuracy: 70.52%
- Final test accuracy: 69.46%
- Rows used after filtering: 28,387
- Train/test split: 22,709 / 5,678

Test accuracy tracked training accuracy closely the whole way through (see mse_curve.png), so no real overfitting. The curve stayed flat until around epoch 3800, then dropped fast, then leveled off classic sigmoid-network training shape.



Artificial Applicants

I tested a strong applicant (GPA 3.95, GRE 332, PhD, international) against an average one (GPA 3.30, GRE 300, Masters, local). The strong applicant got predicted Rejected (31.5% probability), and the average one got predicted Accepted (73.6%) which is backwards from what I'd expected.

My guess is the model is picking up on the fact that PhD and international applicants have lower acceptance rates overall in the dataset, and that categorical signal outweighs the actual GPA/GRE numbers.

What's useful about implementing a neural network manually?

You see what's happening at each step the matrix shapes, the gradients, the update rule instead of it all being hidden inside a library call.

Limitations of MSE for binary classification?

MSE doesn't punish confident wrong answers hard enough, and it wasn't built for classification in the first place cross-entropy would push the model toward better decision boundaries faster.

What's missing from this dataset?

Stuff that matters for admissions like letters of rec, research fit, statement of purpose, funding plus it's self-reported, so who posts their results at all is already biased.